

Niels Vanhasbroeck

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RESEARCH INTEREST

- Mathematical psychology
- Psychometrics
- Emotion, Attitudes, Walking behavior

WORK EXPERIENCE

Post-doctoral Researcher 2025 – present

University of Amsterdam

Validating the Attitudinal Blume-Capel model to capture opinion dynamics.

Post-doctoral Researcher 2023 – 2025

University of Amsterdam

Investigating the validity of the Minds for Mobile Agents pedestrian model and providing software for its simulation and estimation.

PhD Student 2019 – 2023

KU Leuven

Validating the Affective Ising Model, a drift-diffusion model to capture affect dynamics. Supervision by Prof. Dr. Francis Tuerlinckx (supervisor), and Prof. Dr. Wolf Vanpaemel and Prof. Dr. Agnes Moors (co-supervisors).

Tutor in Statistics 2018 – 2019

Rebus Leuven

Teaching basic and advanced statistical principles to psychology students.

EDUCATION

Master of Science, Psychology 2017 – 2019

KU Leuven

Bachelor in Psychology 2014 – 2017

KU Leuven

PUBLICATIONS

Ariens, S., **Vanhasbroeck, N.**, Lafit, G., & Ceulemans, E. (preprint). Interpreting parameters of dynamic regression models [Preprint available at https://osf.io/preprints/psyarxiv/4tznv_v1].

Kandala, R., **Vanhasbroeck, N.**, & Hoemann, K. (preprint). Evaluating BERTopic on open-ended data: A case study with Belgian Dutch daily narratives [Preprint available at <https://arxiv.org/pdf/2504.14707?>].

- Kandala, R., **Vanhasbroeck, N.**, Tamm, S., Van hamme, H., Kuppens, P., Mesquita, B., & Hoemann, K. (preprint). Toward an ecology of emotion in everyday life [Preprint available at https://osf.io/preprints/psyarxiv/hfvnx_v1].
- Vanhasbroeck, N.**, Yu, K., & Ariens, S. (preprint). No need to be indirect: On the role of data in the validation of theory [Preprint available at https://osf.io/preprints/psyarxiv/6f4q7_v1].
- Tuerlinckx, F., Kuppens, P., Ariens, S., Cloos, L., Dejonckheere, E., Lafit, G., Niemeijer, K., Revol, J., Schat, E., Schreuder, M., **Vanhasbroeck, N.**, & Ceulemans, E. (2026). New developments in experience sampling methodology. *British Journal of Mathematical and Statistical Psychology*, 79, 46–65. <https://doi.org/10.1011/bmps.12398>
- Henninger, M., **Vanhasbroeck, N.**, & Tuerlinckx, F. (2025). Affect dynamics or response bias? The relationship between extreme response style and affect dynamics in a controlled experiment. *Psychological Assessment*, 37, 639–655. <https://doi.org/10.1037/pas0001370>
- Yu, K., Lin, T.-Y., Zaman, J., Tuerlinckx, F., & **Vanhasbroeck, N.** (2025). Consistency of perceptual response variability in size estimation and reproduction tasks. *Behavior Research Methods*, 57, 127. <https://doi.org/10.3758/s13428-025-02650-1>
- Vanhasbroeck, N.**, Loossens, T., & Tuerlinckx, F. (2024). Discounting models as a special case of the VARMAX. *Journal of Mathematical Psychology*, 120-121, 102856. <https://doi.org/10.1016/j.jmp.2024.102856>
- Vanhasbroeck, N.**, Niemeijer, K., & Tuerlinckx, F. (2024). Nonlinearity in affect dynamics persists after accounting for the valence of daily-life events. *Emotion*, 24, 1206–1223. <https://doi.org/10.1037/emo0001336>
- Vanhasbroeck, N.**, Vanbelle, S., Moors, A., Vanpaemel, W., & Tuerlinckx, F. (2024). Chasing consistency: On the measurement error in self-reported affect in experiments. *Behavior Research Methods*, 56, 3009–3022. <https://doi.org/10.3758/s13428-023-02290-3>
- Chalkia, A., **Vanhasbroeck, N.**, Van Oudenhove, L., Kindt, M., & Beckers, T. (2023). Emotional associative memory is disrupted by directed forgetting. *Communications Psychology*, 1, 24. <https://doi.org/10.1038/s44271-023-00024-x>
- Tanjitpiyanond, P., Jetten, J., Peters, K., Ashokkumar, A., Barry, O., Billet, M., Becker, M., Booth, R. W., Castro, D., Chinchilla, J., Costantini, G., Dejonckheere, E., Dindins, G., Erbas, Y., Espinosa, A., Finchilescu, G., Gómez, A., González, T., Goto, N., ... Yeung, V. W.-L. (2023). A 32-society investigation of the influence of perceived economic inequality on social class stereotyping. *European Journal of Social Psychology*, 53, 367–382. <https://doi.org/10.1002/ejsp.2908>
- Loossens, T., Meers, K., **Vanhasbroeck, N.**, Anarat, N., Verdonck, S., & Tuerlinckx, F. (2022). Efficient estimation of bounded gradient-drift diffusion models of affect on CPU and GPU. *Behavior Research Methods*, 54, 1428–1443. <https://doi.org/10.3758/s13428-021-01674-7>
- Vanhasbroeck, N.**, Loossens, T., Anarat, N., Ariens, S., Vanpaemel, W., Moors, A., & Tuerlinckx, F. (2022). Stimulus-driven affective change: Evaluating computational models of affect dynamics in conjunction with input. *Affective Science*, 3, 559–576. <https://doi.org/10.1007/s42761-022-00118-5>
- Vanhasbroeck, N.**, Ariens, S., Tuerlinckx, F., & Loossens, T. (2021). Computational models for affect dynamics. In C. H. Waugh & P. Kuppens (Eds.), *Affect dynamics*. Springer. https://doi.org/10.1007/978-3-030-82965-0_10

Vanhasbroeck, N., Devos, L., Pessers, S., Kuppens, P., Vanpaemel, W., Moors, A., & Tuerlinckx, F. (2021). Testing a computational model of subjective well-being: A preregistered replication of rutledge et al. (2014). *Cognition and Emotion*, 35, 822–835. <https://doi.org/10.1080/02699931.2021.1891863>

Chalkia, A., Schroyens, N., Leng, L., **Vanhasbroeck, N.**, Zenses, A.-K., Van Oudenhove, L., & Beckers, T. (2020). No persistent attenuation of fear memories in humans: A registered replication of the reactivation-extinction effect. *Cortex*, 129, 496–509. <https://doi.org/10.1016/j.cortex.2020.04.017>

TALKS AND POSTERS

Henninger, M., **Vanhasbroeck, N.**, & Tuerlinckx, F. (2025). Affect dynamics or response bias? The relationship between extreme response style and affect dynamics in a controlled experiment [Presented instead of the first author at the *Annual Meeting of the Society for Ambulatory Assessment*, Leuven, 26 May 2025 – 28 May 2025].

Vanhasbroeck, N. (2025a). Complex systems in psychology: Applications in affect dynamics and walking behavior [Presented at the weekly seminar of the *Psychometrics and Statistics Group*, University of Groningen, 1 April 2025].

Vanhasbroeck, N. (2025b). Psychology through the lens of complex systems [Presented at a lab meeting of Laura Bringmann's *Intensive Longitudinal Data Lab*, University of Groningen, 22 January 2025].

Vanhasbroeck, N. (2025c). Walking you through the Minds for Mobile Agents pedestrian model [Presented at the weekly seminar of the *Research Group of Quantitative Psychology and Individual Differences*, KU Leuven, 25 March 2025].

Vanhasbroeck, N., Loossens, T., & Tuerlinckx, F. (2025). Seemingly different yet the same: Discounting models as a special case of the autoregressive models [Presented at the *Annual Meeting of the Society for Mathematical Psychology*, Columbus, 25 July 2025 – 28 July 2025].

Vanhasbroeck, N. & Yu, K. (2025). Can we rely on reliable parameter estimates? [Presented at the *International Meeting of the Psychometric Society*, Minneapolis, 14 July 2025 – 18 July 2025].

Vanhasbroeck, N., Blanken, T., Borsboom, D., Matzke, D., & Heathcote, A. (2024). Minds for Mobile Agents (M4MA): A pedestrian model based on psychological principles [Presented at the *Annual Meeting of the Society for Mathematical Psychology*, Tilburg, 19 July 2024 – 22 July 2024].

Vanhasbroeck, N. & Heathcote, A. (2024). Creating a workflow for preprocessing spatial positions [Poster presented at the *International Meeting of the Psychometric Society*, Prague, 16 July 2024 – 19 July 2024].

Vanhasbroeck, N. (2023). Computational models for affect dynamics [Presented at a colloquium of the *Affect, Motivation, & Action Lab*, Leiden University, 15 June 2023].

Vanhasbroeck, N., Vanbelle, S., Vanpaemel, W., Moors, A., & Tuerlinckx, F. (2023). Assessing the consistency of affective responses [Presented at the *International Meeting of the Psychometric Society*, Maryland, 25 July 2023 – 28 July 2023].

Vanhasbroeck, N., Vanpaemel, W., Moors, A., & Tuerlinckx, F. (2023a). The Affective Ising Model: A nonlinear model of affect dynamics [Presented at the *Annual Meeting of the Society for Mathematical Psychology*, Amsterdam, 18 July 2023 – 21 July 2023].

- Vanhasbroeck, N.,** Vanpaemel, W., Moors, A., & Tuerlinckx, F. (2023b). The Affective Ising Model: A nonlinear model of affect dynamics [Presented at the *Interuniversity Graduate School of Psychometrics and Sociometrics Summer Conference 2023*, Tilburg, 8 June 2023 – 9 June 2023].
- Vanhasbroeck, N.** (2022a). Context-driven affect dynamics: Nonlinear affective change in conjunction with input [Presented at the *Annual Meeting of the Society for Ambulatory Assessment*, Virtual, 14 June 2022 - 16 June 2022].
- Vanhasbroeck, N.** (2022b). Reliability of single-item measures of affect within an experimental context [Presented at the weekly seminar of the *Research Group of Quantitative Psychology and Individual Differences*, KU Leuven, 1 March 2022].
- Vanhasbroeck, N.,** Aeyels, J., Vanpaemel, W., Tuerlinckx, F., & Moors, A. (2022). Affective changes in an uncertain context [Presented at the weekly seminar of the *Research Group of Quantitative Psychology and Individual Differences*, KU Leuven, 14 December 2022].
- Vanhasbroeck, N.,** Loossens, T., Anarat, N., Ariens, S., Vanpaemel, W., Moors, A., & Tuerlinckx, F. (2022a). Computational models of affect dynamics: Accounting for nonlinearity and affective reactivity [Poster presented at the *9th Annual Conference of the Society of Affective Sciences (SAS)*, Virtual, 30 March 2022 - 2 April 2022].
- Vanhasbroeck, N.,** Loossens, T., Anarat, N., Ariens, S., Vanpaemel, W., Moors, A., & Tuerlinckx, F. (2022b). Computational models of affect dynamics: Importance of accounting for nonlinearity and reactivity [Poster presented at the *Conference of the International Society for Research on Emotion (ISRE)*, Los Angeles, 15 July 2022 - 18 July 2022].
- Vanhasbroeck, N.,** Loossens, T., Anarat, N., Ariens, S., Vanpaemel, W., Moors, A., & Tuerlinckx, F. (2022c). Stimulus-driven affective change: Nonlinearity of affect dynamics persists after accounting for input [Poster presented at the *Annual Meeting of the Belgian Association for Psychological Sciences (BAPS)*, Leuven, 2 June 2022 - 3 June 2022].
- Vanhasbroeck, N.** (2021a). State transition in the Affective Ising Model [Presented at the weekly seminar of the *Research Group of Quantitative Psychology and Individual Differences*, KU Leuven, 4 March 2021].
- Vanhasbroeck, N.** (2021b). The Affective Ising Model [Presented at a meeting of the *Brain Research of Anxiety Mechanisms Lab*, KU Leuven, 28 April 2021].
- Vanhasbroeck, N.** & Anarat, N. (2020). Gambling and affective change [Presented at the weekly seminar of the *Research Group of Quantitative Psychology and Individual Differences*, KU Leuven, 18 February 2020].
- Vanhasbroeck, N.,** Chalkia, A., & Beckers, T. (2019). Directed forgetting of associative fear memories [Poster presented at the *Annual Meeting of the Belgian Association for Psychology Science (BAPS)*, Liège, Belgium, 14 May 2019 - 15 May 2019].

SOFTWARE

predped	R package that allows the user to estimate and simulate the Minds for Mobile Agents pedestrian model.
denoiser	R package that allows for adding realistic noise to and filtering this noise from location data.
impulseR	R package that computes and visualizes how a system responds when it is given one or more impulses are given.
InputModels.jl	Julia package that allows the user to estimate and simulate the models defined in GradientDiffusion.jl while accounting for a set of external variables.